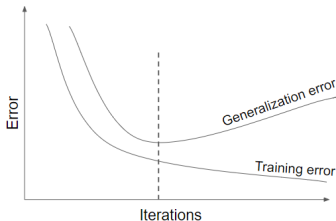


Deep Learning

Early Stopping



Learning goals

- Know how early stopping works
- Understand how early stopping acts as a regularizer
- Know early stopping imitates $L2$ regularization in some cases

EARLY STOPPING

- When training with an iterative optimizer such as SGD, it is commonly the case that, after a certain number of iterations, generalization error begins to increase even though training error continues to decrease.
- **Early stopping** refers to stopping the algorithm early before the generalization error increases.

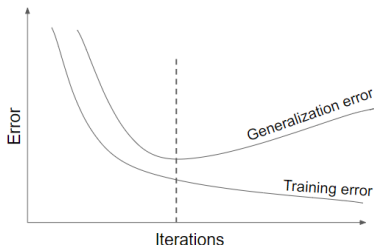


Figure: After a certain number of iterations, the algorithm begins to overfit.

EARLY STOPPING / 2

How early stopping works:

- ➊ Split training data $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{subtrain}}$ and $\mathcal{D}_{\text{val}}$ (e.g. with a ratio of 2:1).
- ➋ Train on $\mathcal{D}_{\text{subtrain}}$ and evaluate model using the validation set $\mathcal{D}_{\text{val}}$.
- ➌ Stop training when validation error stops decreasing (after a range of “patience” steps).
- ➍ Use parameters of the previous step for the actual model.

More sophisticated forms also apply cross-validation.

EARLY STOPPING AND L_2

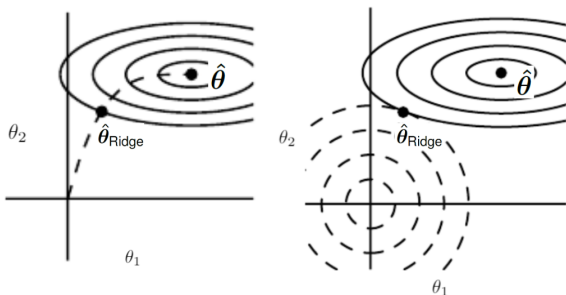
Strengths	Weaknesses
Effective and simple	Periodical evaluation of validation error
Applicable to almost any model without adjustment	Temporary copy of θ (we have to save the whole model each time validation error improves)
Combinable with other regularization methods	Less data for training $\rightarrow$ include $\mathcal{D}_{\text{val}}$ afterwards

- For simple case of LM with squared loss and GD optim initialized at $\theta = 0$: Early stopping has exact correspondence with L_2 regularization/WD: optimal early-stopping iter T_{stop} inversely proportional to λ scaled by step-size α

$$T_{\text{stop}} \approx \frac{1}{\alpha\lambda} \Leftrightarrow \lambda \approx \frac{1}{T_{\text{stop}}\alpha}$$

- Small λ (regu. $\downarrow$) $\Rightarrow$ large T_{stop} (complexity $\uparrow$) and vice versa

EARLY STOPPING AND $L2 / 2$

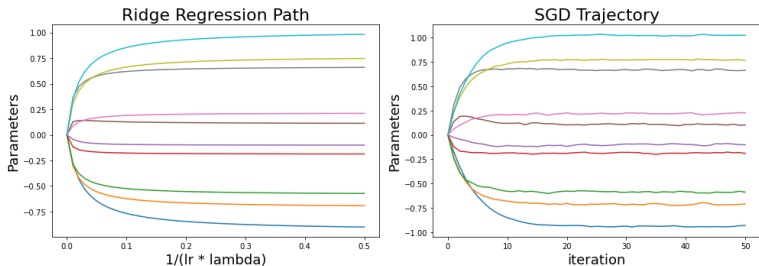


Goodfellow et al., 2016, p. 249 ff.

Figure: Effect of early stopping. *Left:* The solid lines indicate contours of the square loss objective. Dashed line indicates trajectory taken by GD initialized at origin. Instead of reaching minimizer $\hat{\theta}$, ES results in trajectory stopping earlier at $\hat{\theta}_{\text{ridge}}$. *Right:* Effect of $L2$ regularization. Dashed circles indicate contours of $L2$ constraint which push minimizer of regularized cost closer to origin than minimizer of unregularized cost.

SGD TRAJECTORY AND L_2

Solution paths for L_2 regularized linear model closely matches SGD trajectory of unregularized LM initialized at $\theta = 0$



Ali et al., 2020

Caveat: Initialization at the origin is crucial for this equivalence to hold, which is almost never used in practice in ML/DL applications